RMIT Classification: Trusted

Group Project: STABILIZING CONTROL DESIGN

I. OBJECTIVES:

As you have gained sufficient knowledge about control system analysis and design from this course. Thus, you are expected to apply your knowledge to a practical application, particularly rotary inverted pendulum system. The purpose of this project is to: i) understand, assembly, and operate the inverted pendulum dynamic system; ii) run, analyze, and improve the performance of the system.

II. MAIN TASKS (5 TASKS)

Task 1: Mechanical and Electrical Assembly (10%)

Read carefully and understand the user manual [1], especially with understanding about given controller and motor drive. Then, completely install, connect, and assemble to have finished system. For more information and instruction, you can get access to the website [2].

Expected results of task 1:

- A completion of installment and assembly is expected to be done.
- The system is ready to operate in term of hardware.

Task 2: Controller and PC Communication and Interface (15%)

In this task, you are expected to read and understand the guides in files [5][6] and find out on how to communicate the controller (system) to a personal computer. Then, find out on how to use the "Real Time Control System Workbench" [6] that is used to display real time data within the control loop. The user can configure and adjust the control system in real time and observe system response.

Expected results of task 2

- Successfully communicate the controller to the personal computer.
- Successfully install with required apps for the project.

Task 3: Firmware Instalment and Running Test (15%)

In this task, you are expected to: i) download and install firmware into controller; ii) configurate the devices/ports and so on; iii) startup on Windows (Recommended) or Mac OSX (for downloading of the firmware and more information can be found at the website [4]). In this task, you also implement running the system in default mode and record the inverted pendulum dynamic responses.

Expected results of task 3:

- Successfully install the firmware and startup
- Successfully implement running test with default mode to see its performance and stability.
- Record the inverted pendulum dynamic responses and put into the report for analysis and comparison later.

Task 4: Modeling, Control Design, and Simulation (25 %)

Read the materials in the file in [3]. In this task, you are expected to: i) Understand the mathematical model of the system. Then, i) summarize and provide a block-diagram in details to represent signal flow of the systems; ii) simplify the block diagram to achieve a closed-loop transfer function; and iii) then, implement the control analysis and design of the **rotary inverted-pendulum system** using the **Root Locus Method.**

Given requirements of the performance specifications as follows.

- Settling time (8 seconds)
- No steady-state error
- Asymptotical stability
- Well-tracking with disturbance

a) For implementation of the control analysis and design.

- Does P controller be able to achieve the given performance specifications and stability?
- Does PI controllers be able to achieve the given performance specifications and stability?
- Does PD controller be able to achieve the given performance specifications and stability?
- Does PID controller be able to achieve the given performance specifications and stability?

Hint: For each controller, explain clearly why or how you come up with the parameters and conclusions?

b) With the above achieved controller that achieves the given performance specifications and stability, develop your own simulator (simulation software) by using the Matlab/Simulink; iii) Plot dynamic step responses in time (and estimate the overshoot, steady-state error, settling time).

Expected results of task 4:

- A controller is proposed for achieving the given performance specifications and stability.
- A set of parameters are provided for each controller and its conclusion.
- A simulation software and simulation results for the best proposed controller.

Task 5: Modeling, Control Design, and Simulation (10 %)

In Task 4, you found out the best controller to achieve the given performance specifications and stability. Using the Frequency Response Method to verify the achieved results in Task 5.

Task 6: System Parameter Design and Experimental Study (25 %)

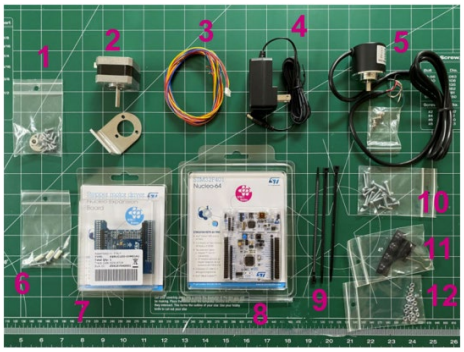
As the system allows students to put a new set of gains (PID) into the control software. Thus, you will use the knowledge to analyze and design a new gain set of PID controller to improve the performance of the system. In this task, you will use gain selection method (called Ziegler-Nichols tuning method) to achieve the new gain set of PID controller. Then, you will run the system with the new set of the gain using the Operation Mode with Modified Parameters. From the operation, you can save time response of the pendulum of two modes (default mode and Modified Parameters mode). With the achieved results, an analysis and comparison of the results would be reported to indicate the performance and stability of the system.

Expected results of task 6:

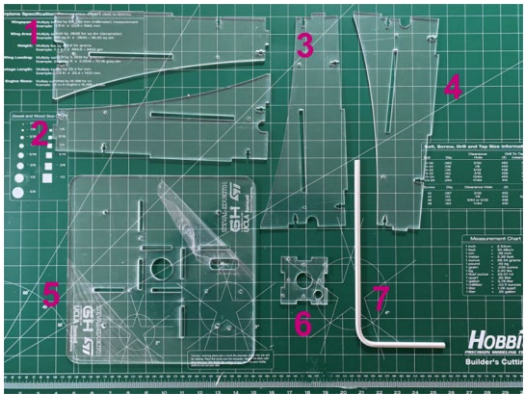
- From your control analysis with real system, the best gain set of PID controllers are expected to be achieved.
- Run the system with the new gain sets and record the time response of the system.

III. PROVIDED MATERIALS AND EQUIPMENT

In this project, you are provided with all components (KIT) of the rotary inverted pendulum dynamic system as below figures. Please, contact Mr. Tai at tai.vo3@rmit.edu.vn for the KIT and signature on paperwork to receive and use the package (KIT) for the project.



1 Motor shaft flange and M3x8 mm machine screws
2 Nema17 stepper motor and encoder L-bracket
3 Motor cables
4 12 V power supply
5 Rotary encoder with M3x8 mm precision screw and washer
6 M3x20 mm machine screw and 10 mm stand-offs
7 X-NUCLEO-IHM01A1 motor driver expansion board
8 NUCLEO-F401RE development board
9 Zip ties
10 M3x16 mm machine screws
11 Adhesive rubber cushions and paper clip
12 M3 machine nuts



1. Acrylic pendulum tower sidewalls
2. Acrylic pendulum tower sidewalls
3. Acrylic pendulum tower sidewalls
4. Acrylic pendulum tower sidewalls
5. Acrylic pendulum bases
6. Acrylic motor mounts
7. Aluminum pendulum rod

IV. REQUIREMENTS

The following are strict requirements of the project.

1. The inverted pendulum dynamic system must be installed entirely from given materials. No other components or materials shall be allowed.
2. We DO NOT have spare parts for replacement. So, if you break the parts or
3. You are expected to work and discuss ONLY on your own group.
4. All components must be uninstalled and return to school after you are done with the project.
5. The project report must follow the Report Template.
6. All sections within the report must be fulfilled. In case it cannot be filled, note the reason why. Make sure you discuss the reason with the Lecturer beforehand, otherwise you may lose points to that portion.
7. The project report should have a high level of detail with S.M.A.R.T. (Specific, Measurable, Achievable, Relevant, Time-based) descriptions.
8. All submitted works must be original. If plagiarism is detected, penalties and disciplinary actions will apply per RMIT Code of Conduct.
9. Submission deadline for the Project Report is **Saturday January 21, 2026, by 11:59 Pm**. The submission is made via Canvas. (One representative will submit on behalf of other members)
10. All project team members are required to attend the **Demonstration session from 12.30-14.00PM on Friday, January 16th, 2026 during your usual Tutorial hours**, in the Robotics Lab, room 8.1.001. Absence from the session means a zero for the testing portion of that student.

V. REFERENCES

[1] assemble document.pfd (you can download from canvas)

[2] https://www.st.com/content/st_com/en/campaigns/educationalplatforms/motorcontrol-edu.html

[3] Introduction_to_Integrated_Rotary_Inverted_Pendulum_v3.pdf

[4] https://sites.google.com/view/ucla-st-motor-control/home

[5] STM32CubeProgrammer_Installation_and_Edukit_Startup_with_Windows.pdf

[6] Edukit_Rotary_Inverted_Pendulum_Manual.pdf

Key contacts and Support:

If you have any question, feel free to contact at:

Lecturer: Thanh Tran at thanh.tran37@rmit.edu.vn

Teaching assistant: Van Le at le.van@rmit.edu.vn

Technician: Mr. Minh Vo minh.vo18@rmit.edu.vn

Other supports (if needed)

If you need to use some special tools or devices in our labs for setting up your project. Please, let Mr. Van or Mr. Minh Vo (minh.vo18@rmit.edu.vn) know what they are. Then, he will check and let you know if your request is available or not. Then, he will inform you for further actions.